Unit 4

Rotational and Circular Motion

Teaching Periods 11 **Weightage %** 8

Amusement Park, Hill park, Karachi

In this unit student should be able to:

- Define angular displacement, Angular Velocity and Angular acceleration and express angular displacement in radians.
- Solve problems by using $S = r\theta$ and $v = r\omega$
- Describe the qualitatively motion in curved path due to perpendicular Force
- Derive and use centripetal acceleration $a = r\omega^2, a = v^2/r$.
- Solve problems using centripetal Force $F = mr\omega^2$, $F = mv^2/r$
- Describe situations in which the centripetal acceleration is caused by tension force, a Frictional force, a gravitational force, or a Normal Force.
- Explain when a vehicle travels round a banked curved at specified speed for the banking angle, the horizontal component of the normal force on vehicle causes the centripetal acceleration.
- Describe the equation $\tan\theta = v^2/rg$, relating banking angle $\theta$to the speed V of vehicle and the radius curvature r.
- Define the term orbital velocity and drive relationship between orbital velocity, gravitational constant, mass and radius of orbit.
- Define the moment of inertia.
- Use the formula of moment of inertia of various bodies for solving problems.
- Define the angular momentum
- Explain the law of conservation of momentum.
- Define the Torque as the cross product of force and moment arm.
- Derive a relation between torque, moment of inertia and angular acceleration.

4.1 Kinematics of Angular Motion

The ***kinematics*** of rotational motion describes the relationships among rotation angle, angular velocity, angular acceleration, and time. Relationship between angle of rotation, angular velocity, and angular acceleration to their equivalents in linear *kinematics.*

we derive all kinematic equations for rotational motion under constant acceleration:

4.1.1 Rotational Quantities

Rotational motion happens when the body itself is spinning. Examples the earth spinning on its axis and the turning shaft of an electric motor. We see a turning wheel motion, but we need a different system of measurement.

There are three basic systems of defining angle measurement.

The revolution is one complete rotation of a body. This unit of measurement of rotational motion is the number of rotations. its unit is the revolution (rev).

Angular Displacement

It is defined as the angle with its vertex at the center of a circle whose sides cut off an arc on the circle equal to its radius figure 4.1. where s = r and θ = 1 rad, then

$\theta = \frac{s}{r}$ …………… (4.1)

Where

θ = angle determined by s and r i.e. angular displacement

s = length of arc of the circle

r = radius of the circle

Angle θ is measured in radians is defined as **the ratio of two lengths: the length of the arc and the radius of a circle.** Since the length units in the ratio cancel, the radian is a dimensionless unit. A useful relationship is 2π rad equals one revolution. Therefore, 1 rev = 360^0 = 2π rad

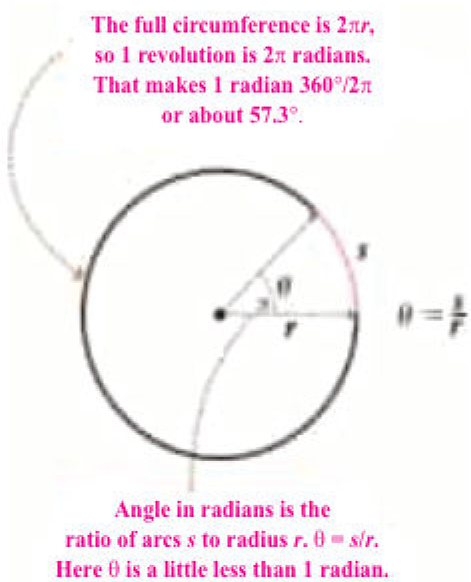

Fig: 4.1

A second system of angular measurement divides the circle of rotation into 360 degrees (360^0 = 1 rev). One degree is $1/360^0$ of a complete revolution.

The radian (rad), which is approximately 57.3^0 or exactly ($360 / 2\pi$), is a third angular unit of measurement.

$$1^o = 2\pi / 360^o = 0.0175 \text{ rad}$$

Angular Velocity

Any object which is rotating about an axis, every point on the object has the same angular velocity. The tangential velocity of any point is proportional to its distance from the axis of rotation.

Angular velocity is the rate of change of angular displacement and denoted by

$\omega_{average} = \frac{\Delta\theta}{\Delta t}$ .... (4.2)

ω = Angular Velocity

θ = Angular Displacement

t = time

This angular velocity is generally measured in **radians per second**, but angles are actually dimensionless. The SI unit for angular velocity is s^{-1} but it usually writes as $rads^{-1}$.

Angular velocity is considered to be a vector quantity, with direction along the axis of rotation in the right – hand rule

The instantaneous angular velocity, which is defined as the instantaneous rate of change of the angular displacement with respect to time, can then be written as

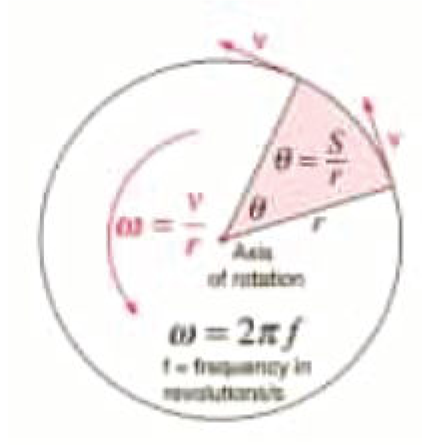

Fig: 4.2

$\omega_{inst} = \lim_{t\to 0}\frac{\Delta\theta}{\Delta t}$ .... (4.3)

The angular velocity is also referred as angular frequency.

$\omega = \frac{2\pi}{t} = 2\pi f$ .... (4.4) As f = 1/t

Another practical unit of angular velocity is revolution per minute (rpm) , usually used in industrial rotatory machines and odometer of vehicles.

$$1\,rpm = \left(\frac{2\pi}{60}\right) rad/s$$

Right -hand rule for angular quantities

Lets consider a circular disk rotating counter clockwise as shown in figure 4.3a, in result the direction of angular velocity vector will be outward direction. In similar way when disk start to rotate clockwise then the direction of angular velocity vector will be inward as represented in figure 4.3b.

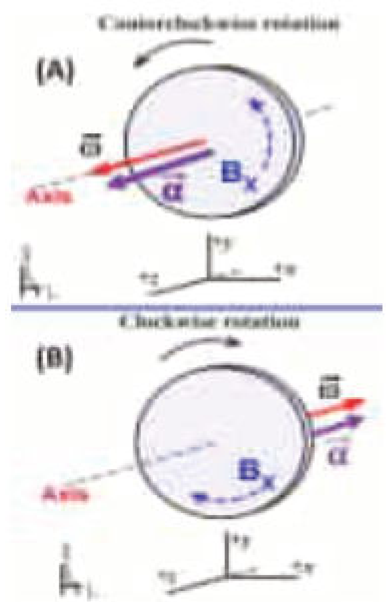

Fig: 4.3

Angular Acceleration

In rotational motion, changing the rate of rotation involves a change in angular velocity and results in an angular acceleration. For uniformly accelerated rotational motion, angular acceleration is defined as the rate of change of angular velocity. That is

$\alpha = \frac{\Delta\omega}{t}$ .... (4.5)

α = Angular acceleration